GE Energy

Power Circuit Breaker Instructions

DS2812B407 500 V dc - 2500 A dc

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Safety Symbol Legend



Indicates a procedure, practice, condition, or statement that, if not strictly observed, could result in personal injury or death.



Indicates a procedure or condition that, if not strictly observed, could result in damage to or destruction of equipment.

Note Indicates an essential or important procedure or statement.

Contents

Section	*Page*
Introduction	3
Receiving and Handling	3
Installation	4
Rating	4
Description	5
Maintenance and Adjustments	10
Latch Adjustment	10
NC Contactor Adjustment	11
Electrical Interlock Adjustment	12
Manual Release Adjustment	13
Padlock Function Check	13
Parts Recommended for Normal Stock	14

Introduction

The breaker is used to apply and remove dc power from an exciter or generator field. The breaker power circuit consists of two normally open (NO) and one normally closed (NC) contacts. The contacts are in a make-before-break configuration. There are 14 convertible auxiliary contacts available. The two NO contacts apply dc power to the field. The NC contact inserts a discharge resistor into the field circuit. The breaker is mechanically latched when closed. There is an anti-pump circuit that will insure that there is only one closure per close command. The breaker is designed for easy access for maintenance purposes. The breaker is a non-draw out type, with a padlock feature to lock the breaker in the open position. With the breaker latched in, there are means for a mechanical as well as an electrical trip. Other relays have been incorporated which provide the anti-pump feature. The breaker consists of two normally open IC28001180 contactors mechanically joined together which, when closed, open the contacts of a normally closed IC2800EA900 contactor. Instruction books *GEH-751E* and *GEH-4286* should be used in conjunction with this instruction book.



This equipment contains a potential hazard of electrical shock or burn. Only personnel who are adequately trained and thoroughly familiar with the equipment and these instructions should install, operate or maintain this equipment.



When using this device for replacement, care should be taken to insure power circuits are connected as shown in Figure 1.

Receiving and Handling

Immediately upon receipt, the equipment should be carefully unpacked and examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly. If the breaker is to be stored, it should be protected from debris, in a dry area.

Installation

The breaker should be carefully inspected before installation to see that parts and adjustments, as indicated in these instructions, are in proper operating condition. Any packing material or other foreign matter should be removed, particularly from the contact tip and armature assemblies.

Grease or other foreign material must be removed from the armature and magnet mating surfaces. Use a cloth dampened with a suitable solvent such as xylol. The armature and magnet should then operate positively and quietly in the closed position. The armature should be closed manually to be sure that the contactor operates without excessive friction or binding.

The breaker *must* be mounted vertically. Care should be taken to insure that the breaker is protected from debris that may jam contacts.

Rating

Maximum continuous dc current rating - NO poles	2500 A
Maximum continuous dc current rating - NC pole	900 A
Maximum continuous dc voltage rating	500 V dc
Maximum interrupting current; also 5 second closing contact current rating	9000 A
Maximum interrupting voltage	2000 V
Close time	0.250 sec
Trip time	0.033 sec
Dc control power	125/250 V dc

Description

The breaker consists of two major circuits, a trip circuit, and a close circuit. There is an anti-pump circuit and a circuit to provide auxiliary contacts, which work in conjunction with the close and trip circuits.

For the following discussion of operation, Figure 1 may be used.

Apply dc control power to the breaker. Place the 41 switch in the close position. Relay 41X will energize through the normally closed contacts of 41Y, 8PD and 41 Auxiliary. Energizing 41X will result in its contacts closing. These contacts are in series with the *close* coils 41CCA and 41CCB. In order for the breaker to mechanically latch, 41CC must remain energized for approximately 250 milliseconds. This is accomplished by 41X being a time delay relay. Relay 41Y will insure only one closure of the main poles per *close* command. The above sequence may be performed by operating the *close* push button located on the face of the breaker. Once the breaker has picked up, it is mechanically latched in. If the dc control power were removed, the breaker will remain latched in. With power removed, the breaker may be tripped by operation of the *trip* push button on the face of the breaker.

Note 41CCA and 41CCB each have two coils that must have opposite polarity so that flux in one coil has a return path through the other. In other words terminal 2 of coils within 41CCA and 41 CCB must be connected with a jumper. 41CCA and 41CCB are then connected in series. Refer to figure 1-A

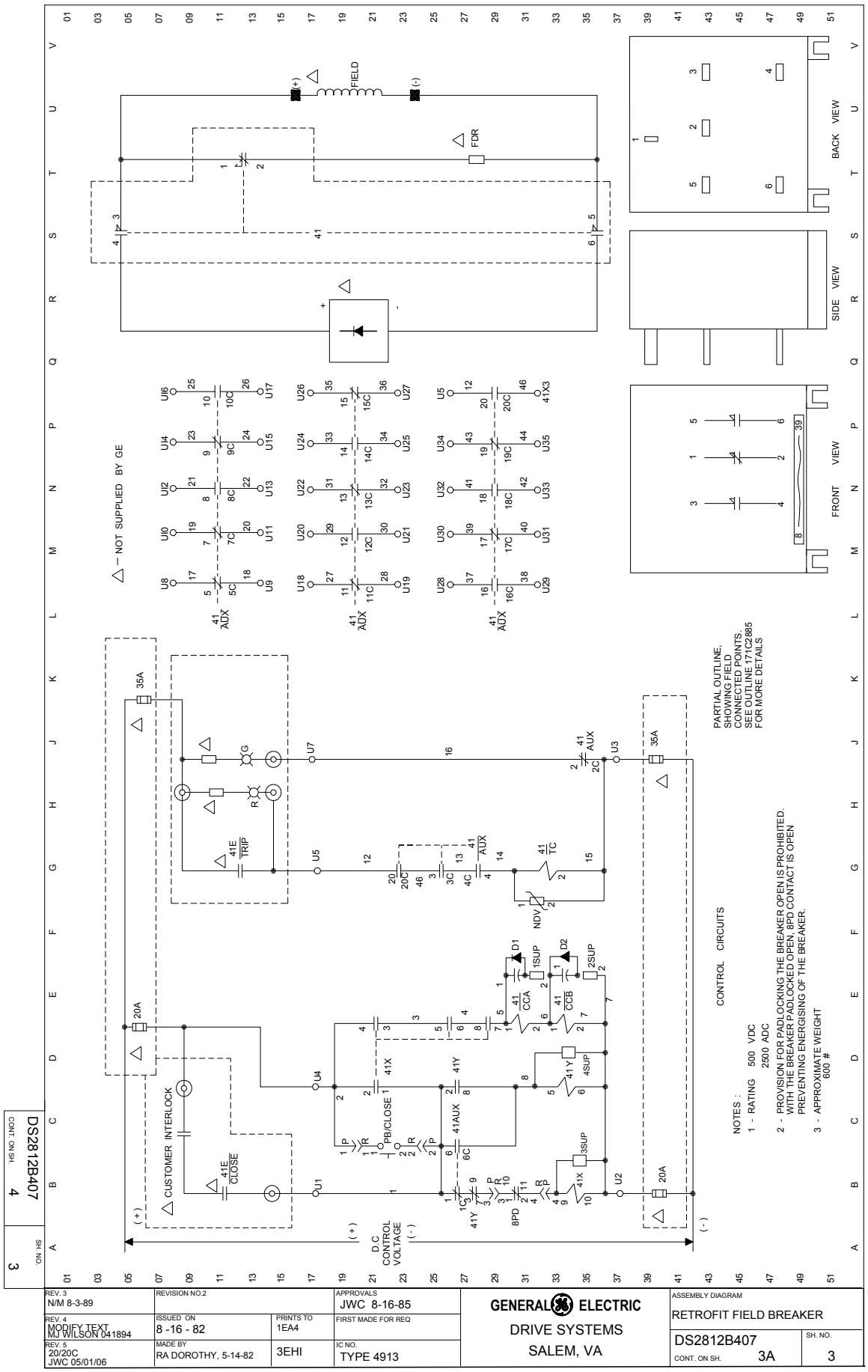
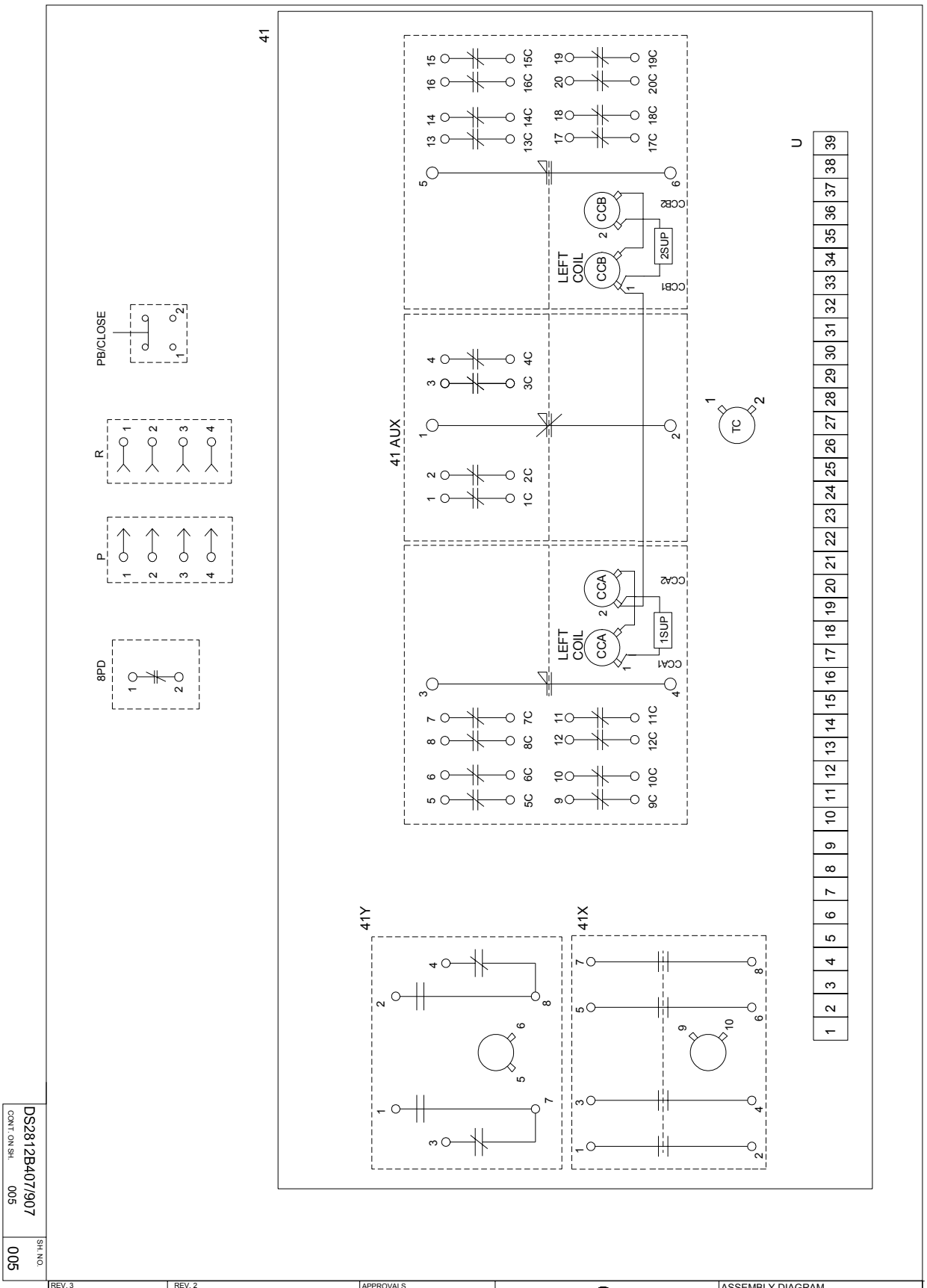


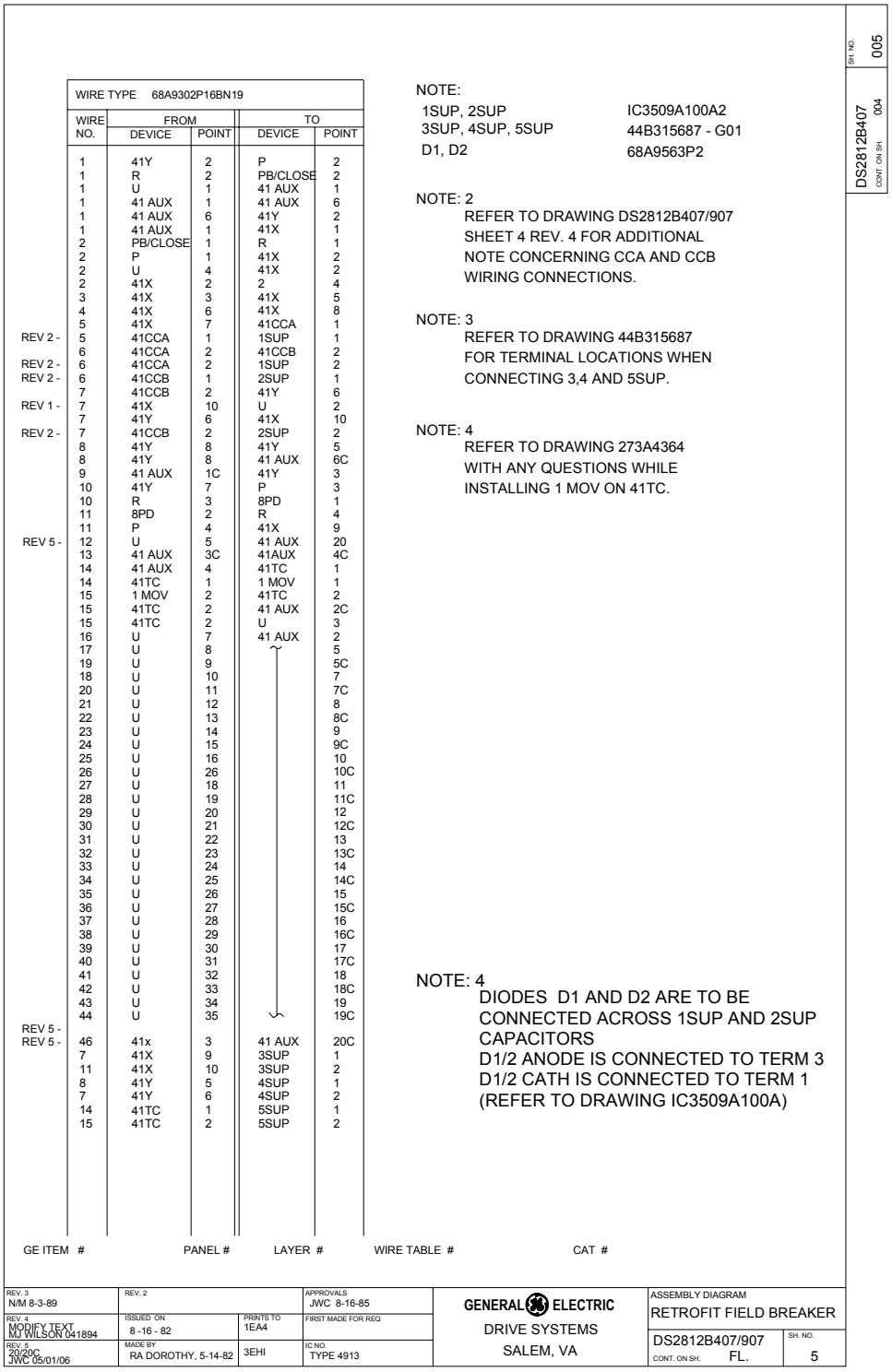
Figure 1 (Sheet 1 of 3)



DS2812B407/907
CONT. ON SH. 005
SH. NO. 005

REV. 3 N/M 8-3-89	REV. 2		APPROVALS JWC 8-16-85	GENERAL ELECTRIC	ASSEMBLY DIAGRAM RETROFIT FIELD BREAKER
REV. 4 MODIFY TEXT MJ WILSON 041894	ISSUED ON 8 -16 - 82	PRINTS TO 1EA4	FIRST MADE FOR REQ	DRIVE SYSTEMS SALEM, VA	DS2812B407/907 CONT. ON SH. 5 / SH. NO. 4
REV. 5 20/20C JWC 05/01/06	MADE BY RA DOROTHY, 5-14-82	3EHI	IC NO. TYPE 4913		

Figure 1 (Sheet 2 of 3)



WIRE TYPE 68A9302P16BN19

	WIRE NO.	FROM DEVICE	FROM POINT	TO DEVICE	TO POINT
	1	41Y	2	P	2
	1	R	2	PB/CLOSE	2
	1	U	1	41 AUX	1
	1	41 AUX	1	41 AUX	6
	1	41 AUX	6	41Y	2
	1	41 AUX	1	41X	1
	2	PB/CLOSE	1	R	1
	2	P	1	41X	2
	2	U	4	41X	2
	2	41X	2	2	4
	3	41X	3	41X	5
	4	41X	6	41X	8
	5	41X	7	41CCA	1
REV 2 -	5	41CCA	1	1SUP	1
	6	41CCA	2	41CCB	2
REV 2 -	6	41CCA	2	1SUP	2
REV 2 -	6	41CCB	1	2SUP	1
	7	41CCB	2	41Y	6
REV 1 -	7	41X	10	U	2
	7	41Y	6	41X	10
REV 2 -	7	41CCB	2	2SUP	2
	8	41Y	8	41Y	5
	8	41Y	8	41 AUX	6C
	9	41 AUX	1C	41Y	3
	10	41Y	7	P	3
	10	R	3	8PD	1
	11	8PD	2	R	4
	11	P	4	41X	9
REV 5 -	12	U	5	41 AUX	20
	13	41 AUX	3C	41AUX	4C
	14	41 AUX	4	41TC	1
	14	41TC	1	1 MOV	1
	15	1 MOV	2	41TC	2
	15	41TC	2	41 AUX	2C
	15	41TC	2	U	3
	16	U	7	41 AUX	2
	17	U	8		5
	19	U	9		5C
	18	U	10		7
	20	U	11		7C
	21	U	12		8
	22	U	13		8C
	23	U	14		9
	24	U	15		9C
	25	U	16		10
	26	U	26		10C
	27	U	18		11
	28	U	19		11C
	29	U	20		12
	30	U	21		12C
	31	U	22		13
	32	U	23		13C
	33	U	24		14
	34	U	25		14C
	35	U	26		15
	36	U	27		15C
	37	U	28		16
	38	U	29		16C
	39	U	30		17
	40	U	31		17C
	41	U	32		18
	42	U	33		18C
	43	U	34		19
	44	U	35		19C
REV 5 -					
REV 5 -	46	41x	3	41 AUX	20C
	7	41X	9	3SUP	1
	11	41X	10	3SUP	2
	8	41Y	5	4SUP	1
	7	41Y	6	4SUP	2
	14	41TC	1	5SUP	1
	15	41TC	2	5SUP	2

NOTE:

1SUP, 2SUP	IC3509A100A2
3SUP, 4SUP, 5SUP	44B315687 - G01
D1, D2	68A9563P2

NOTE: 2
REFER TO DRAWING DS2812B407/907 SHEET 4 REV. 4 FOR ADDITIONAL NOTE CONCERNING CCA AND CCB WIRING CONNECTIONS.

NOTE: 3
REFER TO DRAWING 44B315687 FOR TERMINAL LOCATIONS WHEN CONNECTING 3,4 AND 5SUP.

NOTE: 4
REFER TO DRAWING 273A4364 WITH ANY QUESTIONS WHILE INSTALLING 1 MOV ON 41TC.

NOTE: 4
DIODES D1 AND D2 ARE TO BE CONNECTED ACROSS 1SUP AND 2SUP CAPACITORS
D1/2 ANODE IS CONNECTED TO TERM 3
D1/2 CATH IS CONNECTED TO TERM 1
(REFER TO DRAWING IC3509A100A)

SH. NO. 005
DS2812B407 CONT. ON SH. 004

GE ITEM # PANEL # LAYER # WIRE TABLE # CAT #

REV. 3 N/M 8-3-89	REV. 2		APPROVALS JWC 8-16-85	GENERAL ELECTRIC DRIVE SYSTEMS SALEM, VA	ASSEMBLY DIAGRAM RETROFIT FIELD BREAKER
REV. 4 MODIFY TEXT MJ WILSON 041894	ISSUED ON 8 -16 - 82	PRINTS TO 1EA4	FIRST MADE FOR REQ		DS2812B407/907 CONT. ON SH. FL. / SH. NO. 5
REV. 5 20/20C JWC 05/01/06	MADE BY RA DOROTHY, 5-14-82	3EHI	IC NO. TYPE 4913		

Figure 1 (Sheet 3 of 3)

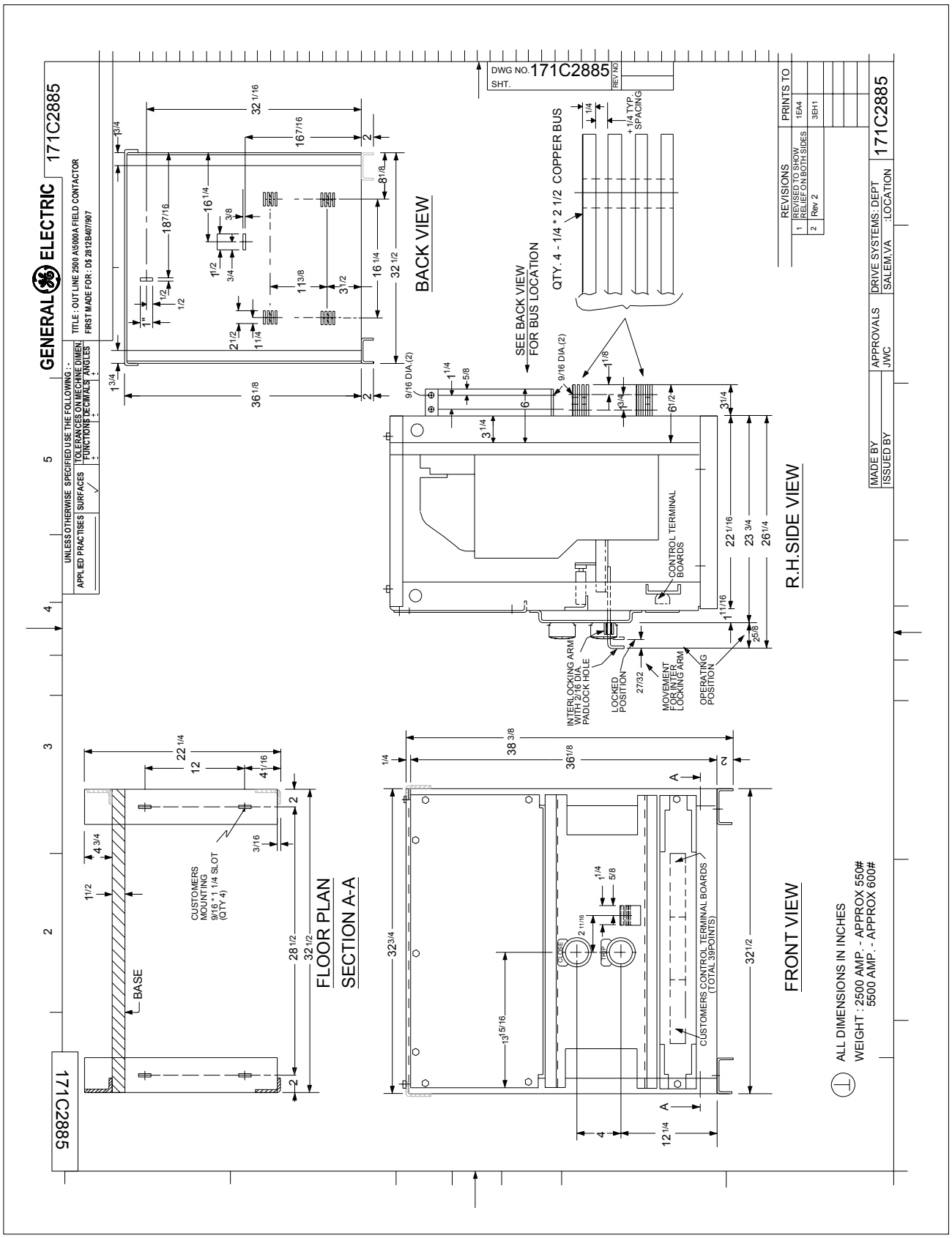

GENERAL ELECTRIC
171C2885
TITLE : OUT LINE 2500 A/5000A FIELD CONTACTOR
FIRST MADE FOR : DS 28 12B407/3907
DWG NO. 171C2885
SHT.
REV NO
BACK VIEW
SEE BACK VIEW FOR BUS LOCATION
QTY. 4 - 1/4 * 2 1/2 COPPER BUS
1/4 TYP SPACING
R.H.SIDE VIEW
CONTROL TERMINAL BOARDS
INTERLOCKING ARM WITH 2 HOLES FOR PADLOCK HOLE
LOCKED POSITION
MOVEMENT FOR INTER LOCKING ARM
OPERATING POSITION
FRONT VIEW
CUSTOMERS CONTROL TERMINAL BOARDS (TOTAL 39POINTS)
FLOOR PLAN
SECTION A-A
BASE
CUSTOMERS MOUNTING 9/16 * 1 1/4 SLOT (QTY 4)
ALL DIMENSIONS IN INCHES
WEIGHT : 2500 AMP. - APPROX 550#
5500 AMP. - APPROX 600#
REVISIONS
PRINTS TO
1 REVISED TO SHOW RELIEF ON BOTH SIDES 1EA4
2 Rev 2 3EH1
MADE BY
ISSUED BY
APPROVALS
JWC
DRIVE SYSTEMS: DEPT
SALEM,VA :LOCATION
171C2885
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING : -
APPLIED PRACTISES
SURFACES
TOLERANCES ON MECHINE DIMEN.
FUNCTIONAL DECIMALS
ANGLES

Figure 2

With dc power applied, place the 41 switch in the *trip* position. The 41 Auxiliary contacts 3-3C, 4-4C, 20-20C will be closed due to the breaker being in the closed position. Closing the 41 switch will allow 41TC to be energized. This action will open the breaker. As stated before, the breaker may be opened by operation of the *trip* push button on the face of the breaker.

Note Version C of this instruction has been modified to add contacts 20-20C in the trip coil circuit in order to improve interruption at 250 V dc. These contacts are no longer wired to terminal board points U36 and U37.

There are 14 convertible contacts wired out for use. These contacts follow the state of the breaker. Points are available for connecting indicating lights.

The contact labeled 8PD is mechanically opened when the breaker is padlocked open. This prevents the breaker from being accidentally energized.

Maintenance and Adjustments

The breaker will provide maximum service if inspected and maintained on a regular basis. The life of the breaker depends upon the severity of its service. Instruction books *GEH-751E* and *GEH-4286* should be used in conjunction with the following adjustment procedure.

Latch Adjustment

Solenoid

Adjust the solenoid armature gap to 0.18/0.25 in. (Latch in tripped position).

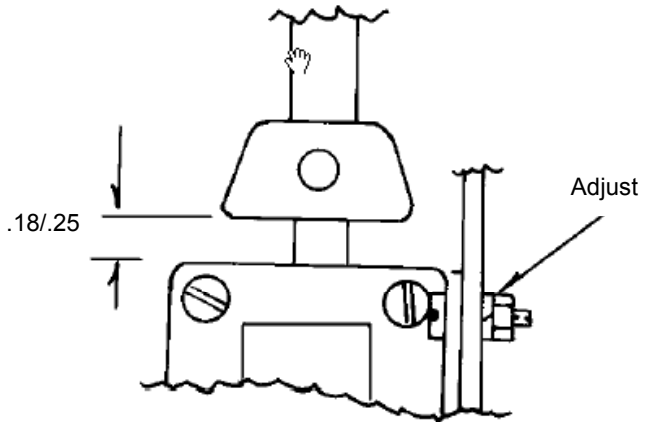


Figure 3

Inspect Latch

Close the IC28001180 contactors and wedge in this position. Manually rotate the latch. No sticking binding or roughness is permitted, due to the fact that needle bearings used on the latch shaft are basically frictionless. Re-adjust latch if required.

NC Contactor Adjustment

Return Springs

Close the IC28001180 contactors and wedge in this position or preferably hold closed at 1/2 rated voltage for positive armature seating. Adjust the NC return springs as shown in the following figure.

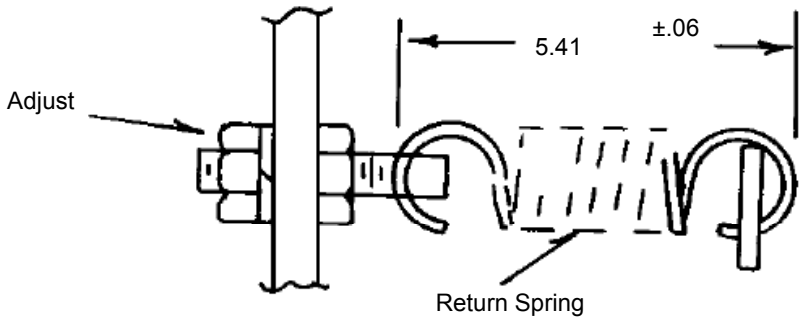


Figure 4

Mechanical Operator

Close the IC28001180 contactors and hold at ½ rated voltage. Adjust the NC contactor operator for 0.010/0.050 stop gap.

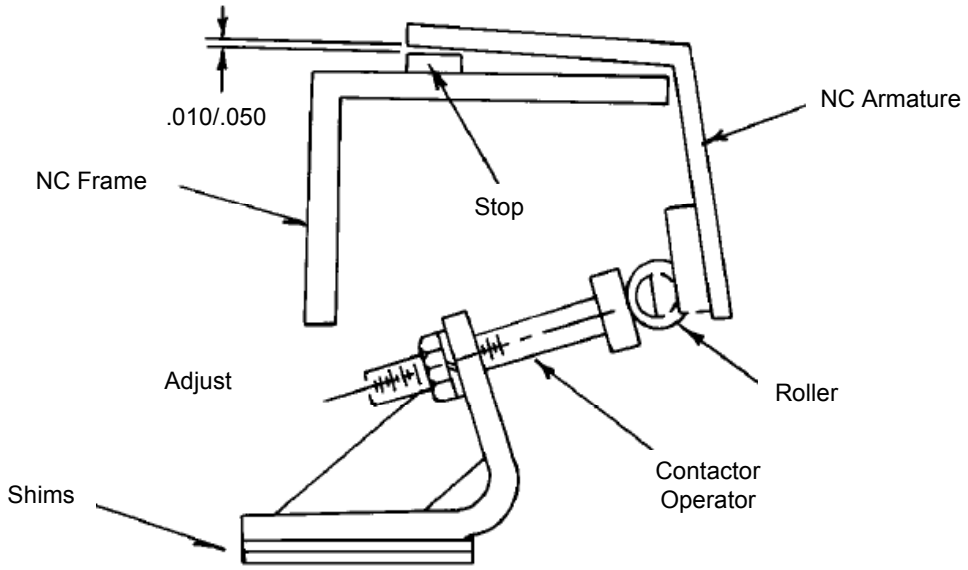


Figure 5

Inspect

Close the IC28001180, NO contactor and hold at ½ rated voltage. Verify that there is some clearance at the NC armature stop. De-energize the IC28001180, NO contactor coils. Keep latch engaged. Verify minimum NC tip gap of 0.42 in. Re-adjust mechanical operator if required.

Electrical Interlock Adjustment

Contactor Operated Interlocks

With the contactor picked up electrically, the auxiliary interlock plungers must have free travel remaining. With the contactor mechanically latched closed, verify minimum auxiliary interlock gap and wipe.

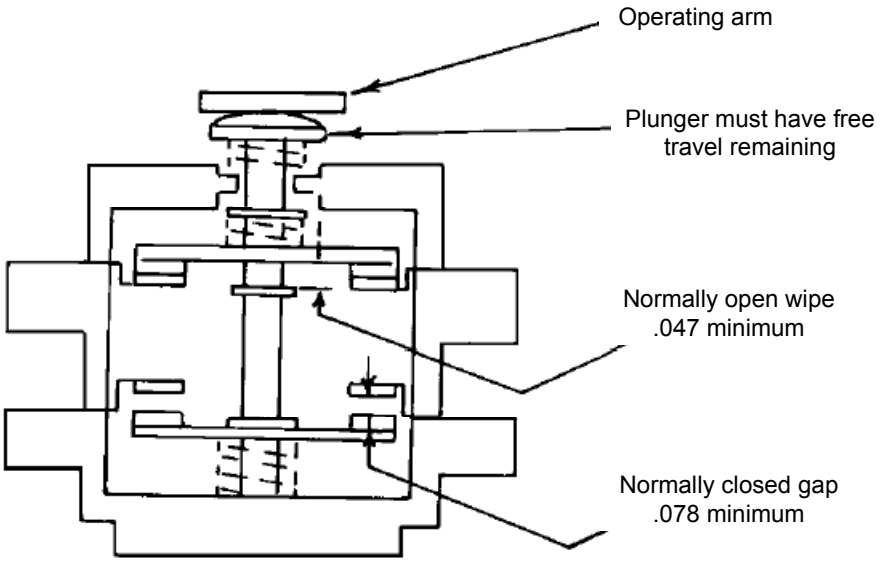


Figure 6

Padlock Interlock

Depress the padlock arm and insert a 0.06 pin in the padlock hole per Figure 8. In this position there must be at least 0.25 clearance between the electrical interlock plunger and the operating arm. Release the padlock arm gently and check for positive interlock operation and minimum wipe per Figure 6. In this case, the interlock plunger may bottom.

Manual Release Adjustment

1 Push the latch (release) and adjust for minimum clearance as shown below.

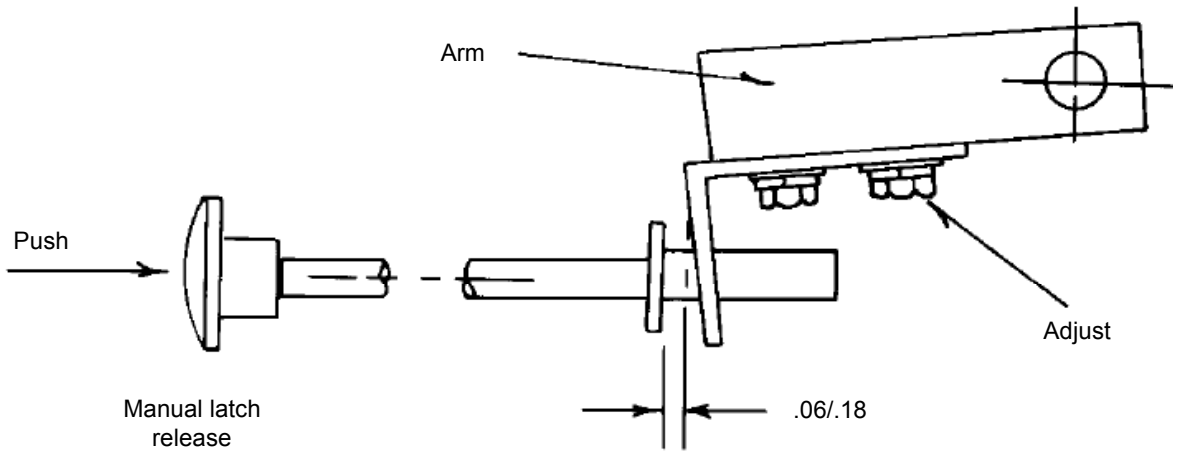


Figure 7

2 Operate at least 10 times by closing electrically and opening manually.

Padlock Function Check

Depress padlock arm to make sure the armature lock rotates above the mechanical joiner. Check that the hole in the padlock arm aligns with the holes in the lock brackets, and that positive armature lock engagement is attained with a 0.06 pin in the padlock hole.

Release the padlock arm. The armature lock should clear the path of mechanical joiner when the joiner assembly is rotated. Repeat depressing and releasing operation several times to assure that no binding occurs in the area of the armature lock.

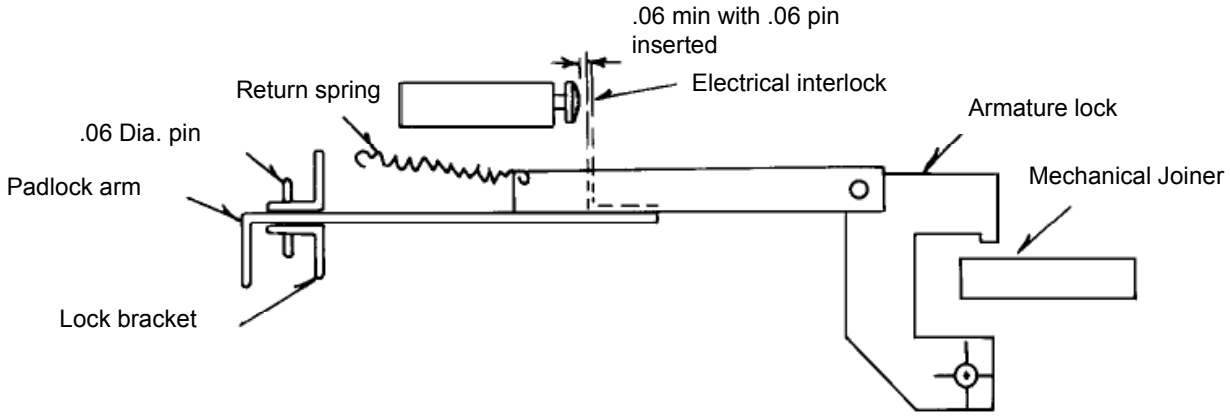


Figure 8

Parts Recommended for Normal Stock

Quantity	Catalog Number	Description
2	173C5480G1	Arc chute assembly
2	2458490	Stationary arcing contact
2	2459875	Movable arcing contact
2	4338327G2	Shunt
2	2413372	Spring for arcing contact
2	9034388G2	Stationary main contact
4	3866756G2	Movable main contact
4	4338327G1	Shunt
4	2413371	Spring for main contact
4	22D63G9A	Coil, 125 V control
2	2D63G17A	Coil, 250 V control
1	117A7313G1	Arc chute assembly
1	259A5562G1	Stationary contact
1	273A2426G1	Movable contact
1	171B3805G2	Shunt
1	259A5572P1	Spring for contact
2	273A4745P1	Return spring
1	DS9500C201AF11A	Solenoid, 125 V control
1	DS9500C201AF10A	Solenoid, 250 V control
1	12HGA11J52	HGA relay, 125 V control
1	12HGA11J51	HGA relay, 250 V control
1	IC2820A100EM203L	Time delay relay,125 V control
1	IC2820A100EM202L	Time delay relay, 250 V control
12	DS2956A205C	1NO, 1NC Universal interlock